Harsh Nileshkumar Mistry

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Education

University of Southern California — *Los Angeles, United States*
M.S. in Engineering Management — *Jan 2025 – Dec 2026 (Expected)*
Coursework: Supply Chain MNGT, Spreadsheet Modeling, Global Operations MNGT

Pandit Deendayal Petroleum University — *Gandhinagar, India*
B.Tech. in Mechanical Engineering — *Aug 2019 – May 2023*
Coursework: Manufacturing Processes, Mechanics of Materials, Optimization Techniques

Skills

Strategic Sourcing & NPI: NPI Procurement, BOM/CBOM Management, RFQs, Supplier Evaluation, Cost Analysis
Manufacturing Knowledge: Sheet-Metal Fabrication, Welding, DFM, APQP/PPAP, DFMEA, RCA, Lean Six Sigma
Tools & AI: MCP, Advanced Excel, PowerBi, SQL, Python, PowerShell, Arena PLM, SAP S/4HANA, SolidWorks, R

Work Experience

Generac — *Santa Monica, United States*
Supply Chain Intern — *June 2026 – Aug 2026*
- Supported **NPI sourcing for PWRmicro** by reconciling engineering BOMs and CBOMs, part numbers, annual demand, supplier quotes, and cost targets across **20+ priority components**; identified a **$1.2M annualized sourcing opportunity** and translated cost gaps into RFQ and negotiation priorities
- Coordinated suppliers, engineering, and sourcing to close missing specifications, drawings, reliability reports, quotes, and **change-control** data; maintained traceable component and sourcing records in **Arena PLM and SharePoint**
- Automated **BOM, quote, and document reconciliation** using advanced Excel and PowerShell; flagged quantity, part-number, and pricing mismatches, assessed alternates, and accelerated evidence-based sourcing decisions

USC Bookstore - Distribution Centre — *Los Angeles, United States*
Supply Chain Operations Assistant — *June 2025 – Aug 2025*
- Directed end-to-end **inventory control** for **1000+ SKUs,** introducing **Lean Six Sigma** workflows, **ERP-driven cycle counting,** and **variance auditing** that improved accuracy above **98%** and eliminated recurring bottlenecks
- Designed and deployed automated **reconciliation tools** and variance detection systems in **MS Excel**, cutting restocking cycle time by **15%** and reducing manual errors by **10%**
- Collaborated with warehouse supervisors to **re-map stock flow layouts** using **process mapping** and **slotting analysis, piloting a shelf navigation system** that increased picking accuracy and reduced fulfillment delays

Vindeep Expo Engineers — *Vapi, India*
Supply Chain Analyst — *Sept 2023 – May 2024*
- Directed multi-site **statutory testing operations** for pressure vessels and lifting equipment, creating detailed **project schedules, risk registers, and compliance frameworks** ensuring **100%** regulatory adherence
- Engineered a centralized **compliance & reporting dashboard** (MS Project + Excel) that automated data consolidation, reduced audit prep time by **15%,** and improved traceability for regulatory agencies
- Applied **Kaizen, DFM, APQP/PPAP and PDCA** methodologies to testing workflows, unlocking a **20% efficiency gain** and reducing equipment downtime

Larsen & Toubro — *Surat, Gujarat*
Project Management Intern — *Dec 2022 – May 2023*
- Engineered a portable **robotic welding seam-monitoring system** integrating **SolidWorks prototypes, electrical subsystems, and Python automation,** improving defect detection and process reliability by **20%**
- Conducted **supplier evaluation, cost estimation, and feasibility modeling** using Excel-based cost-benefit and sensitivity analysis, reducing projected costs by **10%**
- Coordinated cross-functional teams using **Agile project management,** ensuring on-time delivery

Banco Products (India) Ltd. — *Vadodara, India*
Quality and Operations Intern — *May 2022 – July 2022*
- Conducted comprehensive **process audits** using **RCA, FMEA, SPC, and Lean tools,** identifying **10+** bottlenecks and improving workflow performance while maintaining ISO 9001 standards
- Led a **welding cost optimization** initiative using **Kaizen and 5S,** achieving a **12%** reduction in welding-related costs

Project Work

AI Supply Chain Decision Workbench (Python, MCP) — *May 2026 – June 2026*
- Built a working **Python Model Context Protocol server** for agentic operations analysis, exposing four governed tools over structured plant data for KPI segmentation, evidence retrieval, opportunity sizing, and recommendation generation; designed for **Claude-compatible clients** with documented **SQL/ERP back-end extension points**
- Analyzed a simulated **12-plant manufacturing network**, prioritized cost and operational risks, and generated a traceable work-stream brief with reusable tool schemas and handoff documentation for nontechnical users

Inventory Policy Optimization (FOQ vs LFL vs POQ) — *Aug 2025 – Nov 2025*
- Built a demand-driven **inventory optimization model** comparing **FOQ, LFL, and POQ** rules using forecasting outputs, cost parameters, and replenishment constraints to evaluate cost-to-serve performance and service levels
- **Modeled total operational cost** by integrating holding, ordering, and back-order penalty components, identifying the replenishment strategy that delivered an **8–15% reduction** in overall inventory cost while maintaining target fill rates
- Used **Excel Solver, R forecasting models, and scenario-based** sensitivity analysis to evaluate parameter trade-offs, enabling data-backed recommendations for inventory policy selection and planning parameter tuning